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COMPUTER SCIENCE & ENGINEERING

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Experiment 2

Name : Arjun Dubey

UID : 23BCS13190

Branch: BE-CSE

Section/Group: KRG-3B

Semester: 6th

Date of Performance: 19-01-2026

Subject Name: Full Stack-II

Subject Code: 23CSH-309

1. Aim: The aim of this implementation is to develop a secure navigation system using Contextbased authentication along with route protection, ensuring that only logged-in users can access authorized pages of the application.

2. Objective:-

- To centrally manage authentication state across the entire application
- To enable secure navigation using protected and public routes
- To ensure only authenticated users can access restricted pages
- To automatically redirect unauthenticated users to the login page while maintaining a scalable code structure

3. Implementation/Code:

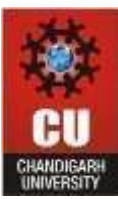
- **Header.jsx :-**

```
import { Link, useNavigate } from "react-router-dom";
import { useAuth } from "../context/AuthContext.jsx";
```

```
const Header = ({ title }) => {
  const { setIsAuthenticated } = useAuth();
  const navigate = useNavigate();
```

```
  const handleLogOut = () => {
    setIsAuthenticated(false);
    navigate("/login");
  };
};
```

```
return (
  <header
```



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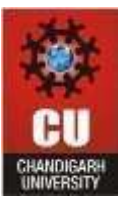
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```
style={{
  backgroundColor: "#ffffff",
  borderBottom: "1px solid #e5e7eb",
  padding: "0.9rem 1.8rem",
  display: "flex",
  alignItems: "center",
  justifyContent: "space-between",
}}
>
{/* App Name */}
<h1
  style={{
    fontSize: "1.4rem",
    fontWeight: "600",
    color: "#065f46",
    margin: 0,
  }}
>
  {title}
</h1>

{/* Links */}
<nav style={{ display: "flex", gap: "1.4rem" }}>
  <Link
    to="/"
    style={{
      textDecoration: "none",
      color: "#374151",
      fontSize: "0.95rem",
    }}
  >
    Dashboard
  </Link>

  <Link
    to="/log"
    style={{
      textDecoration: "none",
      color: "#374151",
      fontSize: "0.95rem",
    }}
  >
    Logs
```



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</Link>

</nav>

/ Logout */*

<button

onClick={handleLogOut}

style={{

backgroundColor: "transparent",

border: "1px solid #dc2626",

color: "#dc2626",

padding: "0.35rem 0.9rem",

borderRadius: "4px",

fontSize: "0.85rem",

cursor: "pointer",

}}

>

Logout

</button>

</header>

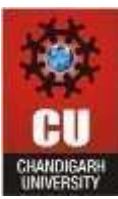
);

};

export default Header;

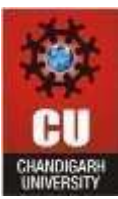
- AuthContext.jsx :-

```
1 import { createContext, useContext, useState } from "react";
2
3 export const AuthContext = createContext(null);
4
5 export const AuthProvider = ({ children }) => {
6   const [isAuthenticated, setIsAuthenticated] = useState(false);
7
8   return (
9     <AuthContext.Provider value={{ isAuthenticated, setIsAuthenticated }}>
10       {children}
11     </AuthContext.Provider>
12   );
13 };
14
15 export const useAuth = () => useContext(AuthContext);
```



- **Login.jsx :-**

```
• import { useAuth } from "../context/AuthContext.jsx";
• import { useNavigate } from "react-router-dom";
•
• const Login = () => {
•   const { setIsAuthenticated } = useAuth();
•   const navigate = useNavigate();
•
•   const handleLogin = () => {
•     setIsAuthenticated(true);
•     navigate("/");
•   };
•
•   return (
•     <div
•       style={{
•         minHeight: "100vh",
•         backgroundColor: "#eef2f7",
•         display: "flex",
•         alignItems: "center",
•         justifyContent: "center",
•       }}
•     >
•       <div
•         style={{
•           width: "380px",
•           backgroundColor: "ffffff",
•           border: "1px solid #e5e7eb",
•           padding: "2rem",
•         }}
•       >
•         <h2
•           style={{
•             fontSize: "1.6rem",
•             fontWeight: "600",
•             color: "#065f46",
•             marginBottom: "0.5rem",
•           }}
•         >
•           Login
•         </h2>
•
•         <p
•           style={{
```



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```
•         fontSize: "0.9rem",
•         color: "#4b5563",
•         marginBottom: "1.5rem",
•     }}
•
•     >
•     Access your EcoTrack dashboard
• </p>
•
• <button
•     onClick={handleLogin}
•     style={{
•         width: "100%",
•         padding: "0.7rem",
•         backgroundColor: "#065f46",
•         color: "#ffffff",
•         border: "none",
•         fontSize: "0.95rem",
•         cursor: "pointer",
•     }}
• >
•     Sign In
• </button>
•
• <p
•     style={{
•         marginTop: "1.3rem",
•         fontSize: "0.8rem",
•         color: "#6b7280",
•     }}
• >
•     Simple authentication demo
• </p>
• </div>
• </div>
• );
• };
•
• export default Login;
•
```

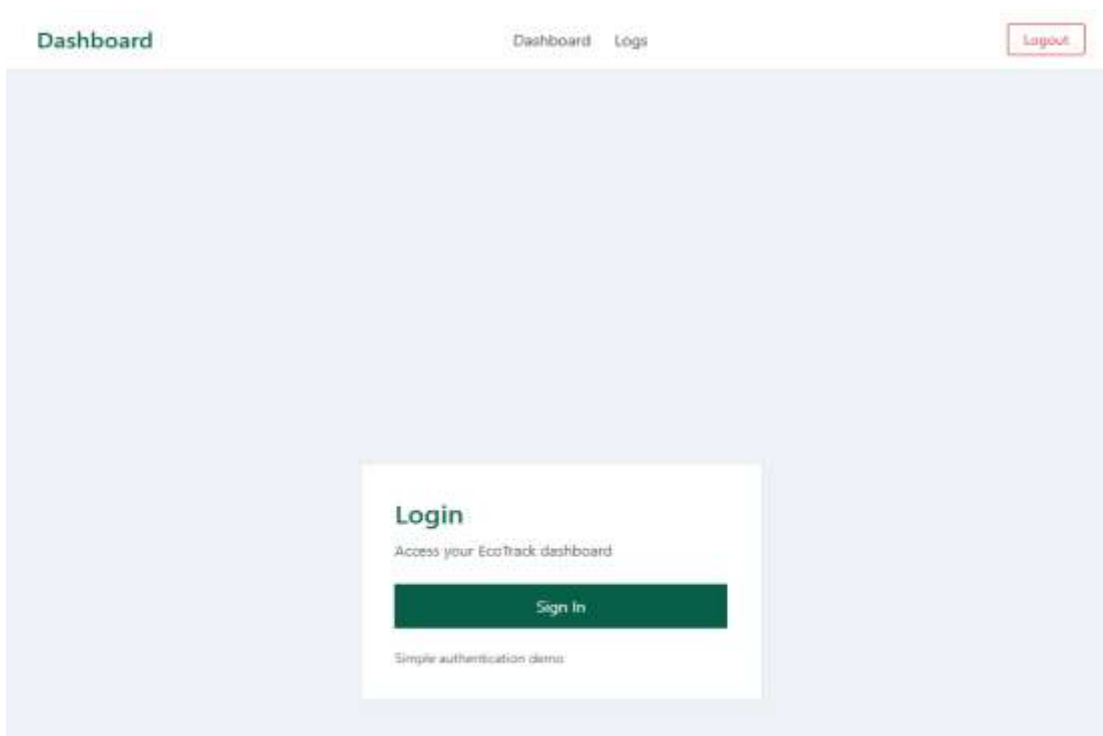
- App.jsx:-

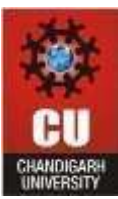
```
1 import Dashboard from "../pages/dashboard.jsx";
2 import Log from "../pages/log.jsx";
3 import Header from "../components/Header.jsx";
4 import { Routes, Route, Link, BrowserRouter } from "react-router-dom";
5 import Login from "../pages/login.jsx";
6 import ProtectedRoute from "../Routes/ProtectedRoute.jsx";
7 import DashboardLayout from "../pages/dashboardLayout.jsx";
8
9 function App() {
10   return (
11     <>
12       <BrowserRouter>
13         <Header title="Dashboard" />
14         <main>
15           <Routes>
16             <Route path="/login" element={ <Login /> } />
17             <Route
18               path="/dashboardLayout"
19               element={
20                 <ProtectedRoute>
21                   <DashboardLayout />
22                 </ProtectedRoute>
23               }
24             />
25             <Route
26               path="/"
27               element={
28                 <ProtectedRoute>
29                   <Dashboard />
30                 </ProtectedRoute>
31               }
32             />
33             <Route
34               path="/log"
35               element={
36                 <ProtectedRoute>
37                   <Log />
38                 </ProtectedRoute>
39               }
40             />
41           </Routes>
42         </main>
43       </BrowserRouter>
44     </>
45   );
46 }
47
48 export default App;
49
```

- ProtectRoute.jsx

```
1 import {Navigate} from "react-router-dom";
2 import {useAuth} from "../context/AuthContext";
3
4 const ProtectedRoute = ({children}) => {
5   const {isAuthenticated} = useAuth();
6
7   if(!isAuthenticated){
8     return <Navigate to="/login" replace />;
9   }
10  return children;
11 }
12 export default ProtectedRoute;
```

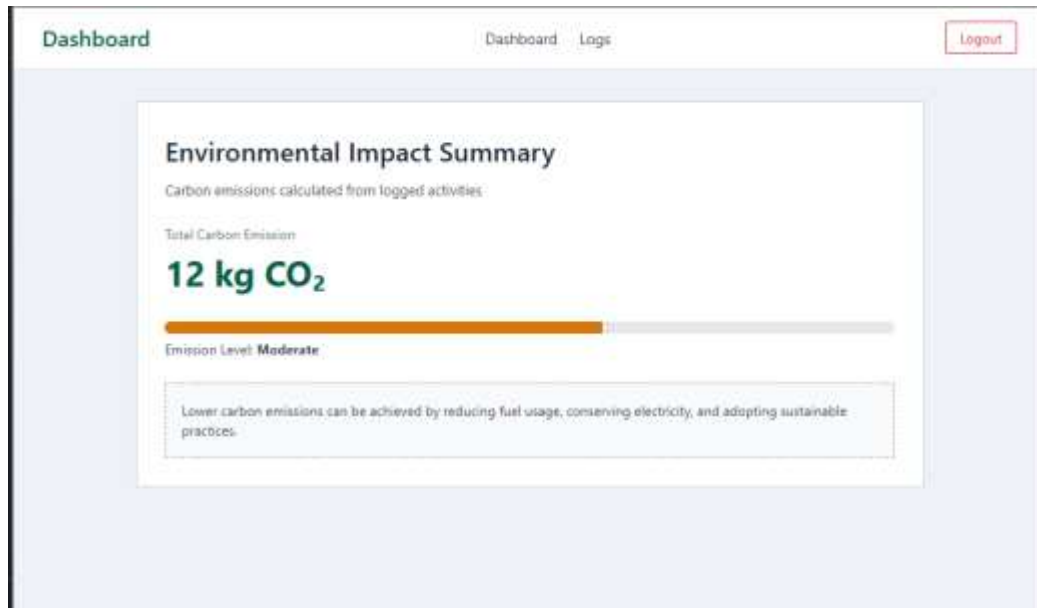
4. Output





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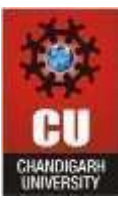
Dashboard [Dashboard](#) [Logs](#) [Logout](#)

Activity Logs

Track your activities and their carbon impact

Activity	Carbon Emission (kg CO ₂)
Car Travel	4
Electricity Usage	6
Public Transport	2

Tip: Choose sustainable options like biking, walking, or public transport to lower your emissions.



5. Learning Outcome :-

- Learned global state management using Context
- Implemented protected routing for secure navigation
- Controlled page access based on authentication status
- Enabled automatic redirection for unauthenticated users
- Improved application security and code maintainability
- Enhanced user experience through seamless navigation
- Built a scalable and reusable authentication structure